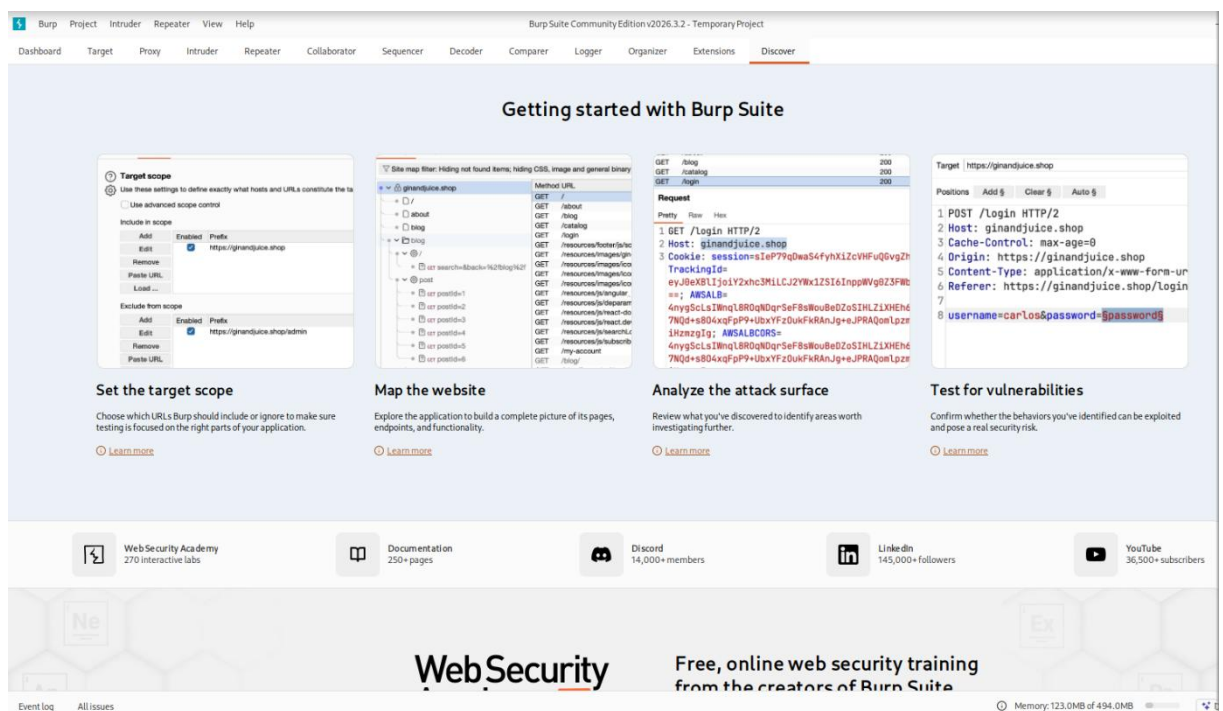


# Sniffing and Evasion

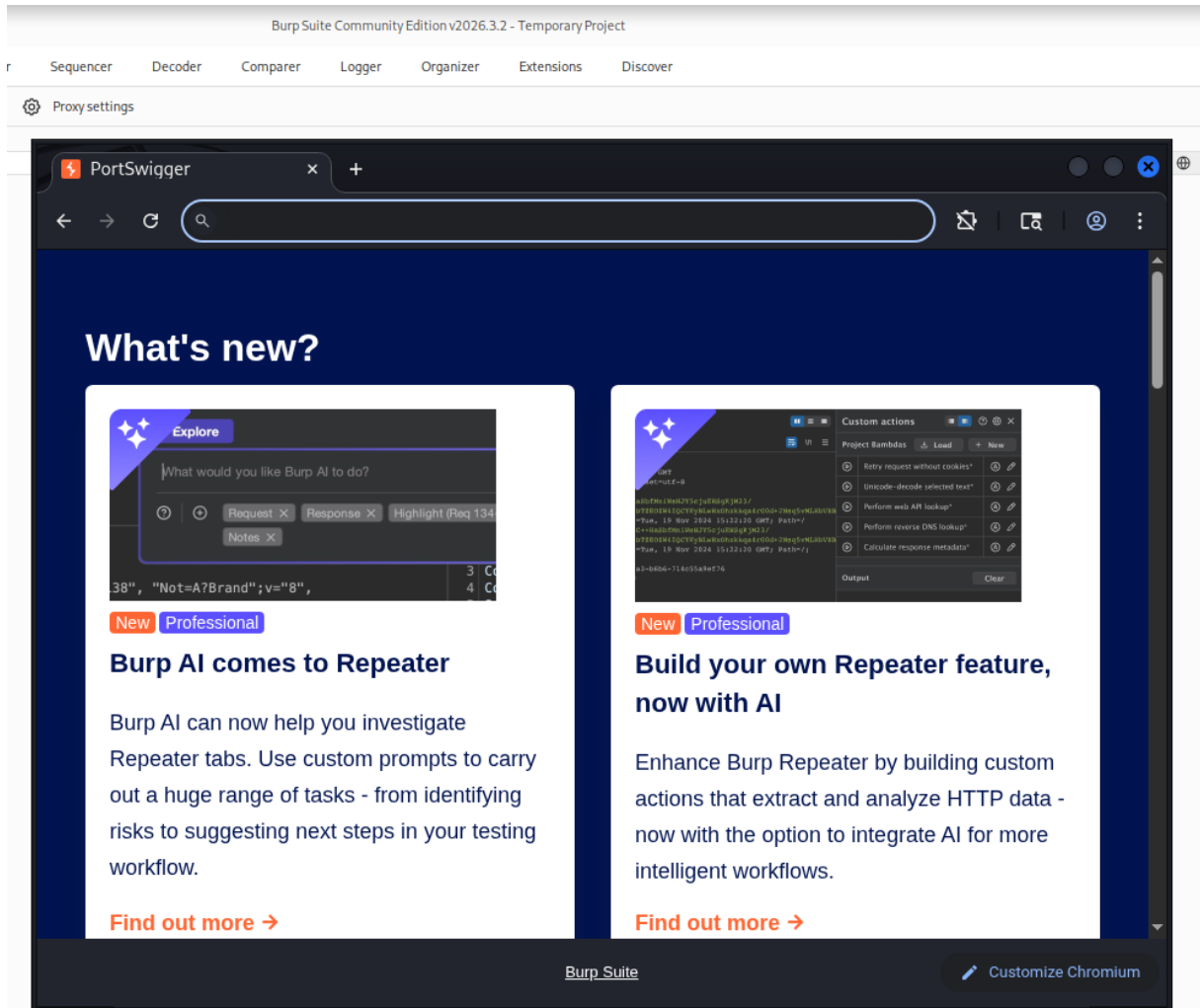
**Objective:** The objective of this exercise was to develop practical web application security testing skills using Burp Suite Community Edition. The exercise focused on understanding how web traffic flows between a browser and a web server, capturing and analysing HTTP requests and responses, intercepting and modifying web requests, examining authentication mechanisms, and using Burp Suite tools such as Proxy, HTTP History, Repeater, and Decoder. The activity also provided hands-on experience in identifying how user input, session information, and server responses are transmitted across web applications, reinforcing key concepts used in web application security assessments and penetration testing.

## TASK 1: Start Burp Suite from Kali



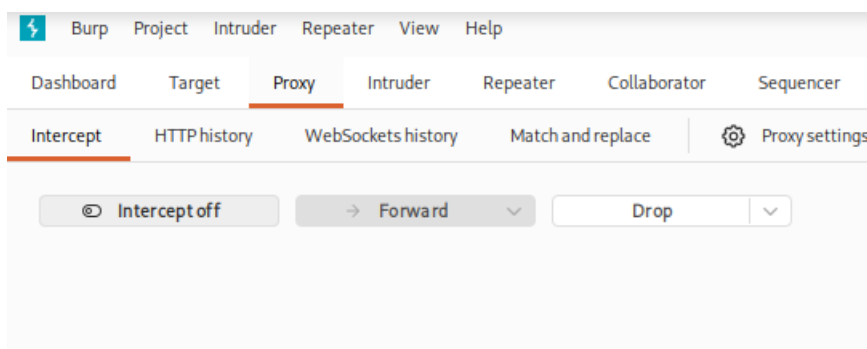
Burp Suite was launched on Kali Linux via Applications → Web Application Analysis → Burp Suite. A temporary project was selected and the default configuration settings were applied. After startup, the main Burp Suite interface loaded successfully, displaying the primary tool tabs such as Dashboard, Target, Proxy, Intruder, and Repeater. This confirmed that Burp Suite was correctly installed and operational in the testing environment.

## TASK 2: Open Burp's Browser



The Proxy tab was accessed and the “Open Browser” option was used to launch Burp’s built-in Chromium browser. This browser was automatically configured to route all traffic through Burp’s local proxy without requiring manual proxy settings. The browser opened successfully, confirming that Burp Suite was actively intercepting-capable and ready to monitor HTTP traffic generated from web browsing activities.

## TASK 3: Check Intercept Status



The Intercept sub-tab under Proxy was reviewed to check interception status. The Intercept feature was initially set to OFF (grey), allowing traffic to pass through Burp Suite without interruption. This configuration ensured that all HTTP requests would still be recorded in the background via HTTP History while normal browsing was maintained without manual request interruption.

## TASK 4a: Visit a Test Website

## TASK 4b: Turn on Intercept and Capture a Request

The screenshot displays the Burp Suite Proxy Intercept tab. At the top, there are three buttons: 'Intercept on' (highlighted in blue), 'Forward' (highlighted in orange), and 'Drop' (grey). To the right of these buttons, it says 'Request to http://testp'. Below the buttons is a table with the following columns: Time, Type, Direction, Method, and URL. The table contains one row: 10:04:44.7 Jun..., HTTP, → Request, GET, http://testphp.vulnweb.com/. Below the table, there is a section titled 'Request' with three tabs: 'Pretty' (selected), 'Raw', and 'Hex'. The 'Pretty' tab shows the following request details:

```
1 GET / HTTP/1.1
2 Host: testphp.vulnweb.com
3 Accept-Language: en-US,en;q=0.9
4 Upgrade-Insecure-Requests: 1
5 User-Agent: Mozilla/5.0 (X11; Linux x86_64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/146.0.0.0 Safari/537.36
6 Sec-Purpose: prefetch;prerender
7 Accept: text/html,application/xhtml+xml,application/xml;q=0.9,image/avif,image/webp,image/apng,*/*;q=0.8,application/signed-exchange;v=b3;q=0.7
8 Accept-Encoding: gzip, deflate, br
9 Connection: keep-alive
10
11
```

Using Burp's embedded browser, the DVWA web application (<http://127.0.0.1/dvwa>) was accessed successfully. The homepage of DVWA loaded, confirming proper communication between the browser, Burp proxy, and the local Apache server hosting the application. This step verified that the test environment was correctly configured for web traffic analysis and security testing activities.

## TASK 5: Forward and Drop Explained

The screenshot displays the Burp Suite interface. At the top, the 'Intercept' tab is active, showing a table of HTTP history. Below this, the 'Request' details panel is open, showing the raw HTTP request data.

| Time                | Type | Direction | Method | URL                                |
|---------------------|------|-----------|--------|------------------------------------|
| 11:38:38 7 Jun 2... | HTTP | → Request | GET    | http://127.0.0.1/dvwa/security.php |

**Request**

Pretty Raw Hex

```
4 sec-ch-ua-mobile: 70
5 sec-ch-ua-platform: "Linux"
6 Accept-Language: en-US,en;q=0.9
7 Upgrade-Insecure-Requests: 1
8 User-Agent: Mozilla/5.0 (X11; Linux x86_64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/146.0.0.0 Safari/537.36
9 Accept: text/html,application/xhtml+xml,application/xml;q=0.9,image/avif,image/webp,image/apng,*/*;q=0.8,application/signed-exchange;v=b3;q=0.7
10 Sec-Fetch-Site: same-origin
11 Sec-Fetch-Mode: navigate
12 Sec-Fetch-User: 71
13 Sec-Fetch-Dest: document
14 Referer: http://127.0.0.1/dvwa/index.php
15 Accept-Encoding: gzip, deflate, br
16 Cookie: security=impossible; PHPSESSID=5df010ef659fd9759ccb0d618093cf3
17 Connection: keep-alive
18
19
```

In this task, Burp Suite was used to intercept a request and demonstrate how traffic is controlled using the Forward and Drop functions. Intercept was enabled, and a request was captured between the browser and the server. The request was then forwarded using the Forward button, allowing it to reach the web server successfully. After forwarding, Intercept was turned off to restore normal browsing. This showed that Forward allows requests to continue to the server, while Drop would have prevented the request from being sent entirely. The browser was then able to load the requested page normally after the request was forwarded.

## TASK 6: Explore Proxy History

| Dashboard Target <b>Proxy</b> Intruder Repeater Collaborator Sequencer Decoder Comparer Logger Organizer Extensions Discover |                             |        |                |        |        |             |        |           |           |                          |     |       |     |              |
|------------------------------------------------------------------------------------------------------------------------------|-----------------------------|--------|----------------|--------|--------|-------------|--------|-----------|-----------|--------------------------|-----|-------|-----|--------------|
| Intercept <b>HTTP History</b> WebSockets history Match and replace Proxy settings                                            |                             |        |                |        |        |             |        |           |           |                          |     |       |     |              |
| Filter settings: Hiding CSS and image content; hiding specific extensions                                                    |                             |        |                |        |        |             |        |           |           |                          |     |       |     |              |
| Pro version only                                                                                                             |                             |        |                |        |        |             |        |           |           |                          |     |       |     |              |
| #                                                                                                                            | Host                        | Method | URL            | Params | Edited | Status code | Length | MIME type | Extension | Title                    | URL | Notes | TLS | IP           |
| 1                                                                                                                            | http://testphp.vulnweb.com  | GET    | /              |        |        |             |        |           |           |                          |     |       |     | 44.228.249.3 |
| 3                                                                                                                            | http://testphp.vulnweb.com  | GET    | /              |        |        |             |        |           |           |                          |     |       |     | 44.228.249.3 |
| 5                                                                                                                            | http://testphp.vulnweb.com  | GET    | /              |        |        |             |        |           |           |                          |     |       |     | 44.228.249.3 |
| 7                                                                                                                            | http://testphp.vulnweb.com  | GET    | /              |        |        |             |        |           |           |                          |     |       |     | 44.228.249.3 |
| 8                                                                                                                            | http://testphp.vulnweb.com  | GET    | /              |        |        |             |        |           |           |                          |     |       |     | 44.228.249.3 |
| 10                                                                                                                           | http://testphp.vulnweb.com  | GET    | /              |        |        |             |        |           |           |                          |     |       |     | 44.228.249.3 |
| 12                                                                                                                           | http://testphp.vulnweb.com  | GET    | /              |        |        |             |        |           |           |                          |     |       |     | 44.228.249.3 |
| 14                                                                                                                           | http://testphp.vulnweb.com  | GET    | /              |        |        |             |        |           |           |                          |     |       |     | 44.228.249.3 |
| 16                                                                                                                           | https://testphp.vulnweb.com | GET    | /              |        |        |             |        |           |           |                          |     |       |     | 44.228.249.3 |
| 18                                                                                                                           | http://testphp.vulnweb.com  | GET    | /              |        |        |             |        |           |           |                          |     |       |     | 44.228.249.3 |
| 20                                                                                                                           | http://127.0.0.1            | GET    | /dwa           |        |        | 301         | 604    | HTML      |           | 301 Moved Permanently    |     |       |     | 127.0.0.1    |
| 21                                                                                                                           | http://127.0.0.1            | GET    | /dwa/          |        |        | 302         | 677    | HTML      |           |                          |     |       |     | 127.0.0.1    |
| 22                                                                                                                           | http://127.0.0.1            | GET    | /dwa/login.php |        |        | 500         | 295    | HTML      | php       |                          |     |       |     | 127.0.0.1    |
| 23                                                                                                                           | http://127.0.0.1            | GET    | /dwa/          |        |        | 302         | 337    | HTML      |           |                          |     |       |     | 127.0.0.1    |
| 24                                                                                                                           | http://127.0.0.1            | GET    | /dwa/login.php |        |        | 500         | 295    | HTML      | php       |                          |     |       |     | 127.0.0.1    |
| 25                                                                                                                           | http://127.0.0.1            | GET    | /dwa/          |        |        | 302         | 337    | HTML      |           |                          |     |       |     | 127.0.0.1    |
| 26                                                                                                                           | http://127.0.0.1            | GET    | /dwa/login.php |        |        | 500         | 295    | HTML      | php       |                          |     |       |     | 127.0.0.1    |
| 27                                                                                                                           | http://127.0.0.1            | GET    | /dwa/          |        |        | 302         | 337    | HTML      |           |                          |     |       |     | 127.0.0.1    |
| 28                                                                                                                           | http://127.0.0.1            | GET    | /dwa/login.php |        |        | 500         | 295    | HTML      | php       |                          |     |       |     | 127.0.0.1    |
| 29                                                                                                                           | http://127.0.0.1            | GET    | /dwa/          |        |        | 302         | 337    | HTML      |           |                          |     |       |     | 127.0.0.1    |
| 30                                                                                                                           | http://127.0.0.1            | GET    | /dwa/login.php |        |        | 500         | 295    | HTML      | php       |                          |     |       |     | 127.0.0.1    |
| 31                                                                                                                           | http://127.0.0.1            | GET    | /dwa/          |        |        | 302         | 337    | HTML      |           |                          |     |       |     | 127.0.0.1    |
| 32                                                                                                                           | http://127.0.0.1            | GET    | /dwa/login.php |        |        | 500         | 295    | HTML      | php       |                          |     |       |     | 127.0.0.1    |
| 33                                                                                                                           | http://127.0.0.1            | GET    | /dwa/          |        |        | 302         | 337    | HTML      |           |                          |     |       |     | 127.0.0.1    |
| 34                                                                                                                           | http://127.0.0.1            | GET    | /dwa/login.php |        |        | 500         | 295    | HTML      | php       |                          |     |       |     | 127.0.0.1    |
| 35                                                                                                                           | http://127.0.0.1            | GET    | /dwa/setup.php |        |        | 200         | 5047   | HTML      |           | Setup: Damn Vulnerabl... |     |       |     | 127.0.0.1    |
| 43                                                                                                                           | http://127.0.0.1            | POST   | /dwa/setup.php |        |        | 302         | 343    | HTML      | php       |                          |     |       |     | 127.0.0.1    |
| 44                                                                                                                           | http://127.0.0.1            | GET    | /dwa/setup.php |        |        | 200         | 5948   | HTML      | php       | Setup: Damn Vulnerabl... |     |       |     | 127.0.0.1    |
| 45                                                                                                                           | http://127.0.0.1            | GET    | /dwa/login.php |        |        |             |        | HTML      | php       |                          |     |       |     | 127.0.0.1    |

The HTTP History tab in Burp Suite was used to review all captured web traffic while browsing through the Burp browser. Even with Intercept disabled, requests such as loading DVWA pages, login pages, and setup pages were recorded automatically. Each entry displayed the HTTP method, host, URL, and response status code. This confirmed that Burp continuously logs traffic in the background, allowing analysis of all previous requests without needing to intercept them in real time.

## TASK 7a: Turn on Intercept

## TASK 7b: Navigate to Login Page

## TASK 7c: Submit Login and Capture

| Intercept <b>HTTP history</b> WebSockets history Match and replace Proxy settings |      |           |        |                                |
|-----------------------------------------------------------------------------------|------|-----------|--------|--------------------------------|
| <div><div>Interception</div><div>Forward</div><div>Drop</div></div>               |      |           |        |                                |
| Request to htt                                                                    |      |           |        |                                |
| Time                                                                              | Type | Direction | Method | URL                            |
| 12:20:54.7 Jun 2...                                                               | HTTP | → Request | POST   | http://127.0.0.1/dwa/login.php |

| Request                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |     |     |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|-----|
| Pretty                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Raw | Hex |
| <pre>1 POST /dwa/login.php HTTP/1.1 2 Host: 127.0.0.1 3 Content-Length: 88 4 Cache-Control: max-age=0 5 sec-ch-ua: "Not-A.Brand";v="24", "Chromium";v="146" 6 sec-ch-ua-mobile: ?0 7 sec-ch-ua-platform: "Linux" 8 Accept-Language: en-US,en;q=0.9 9 Origin: http://127.0.0.1 10 Content-Type: application/x-www-form-urlencoded 11 Upgrade-Insecure-Requests: 1 12 User-Agent: Mozilla/5.0 (X11; Linux x86_64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/146.0.0.0 Safari/537.36 13 Accept: text/html,application/xhtml+xml,application/xml;q=0.9,image/avif,image/webp,image/apng,*/*;q=0.8,application/signed-exchange;v=b3;q=0.7 14 Sec-Fetch-Site: same-origin 15 Sec-Fetch-Mode: navigate 16 Sec-Fetch-User: ?1 17 Sec-Fetch-Dest: document 18 Referer: http://127.0.0.1/dwa/login.php 19 Accept-Encoding: gzip, deflate, br 20 Cookie: security=impossible; PHPSESSID=5df010ef659fd9759ccbf0d618093cf3 21 Connection: keep-alive 22 23 username=admin&amp;password=password&amp;Login&gt;Login&amp;user_token=33ecf57577e94fdaf78a315eebb4dc27</pre> |     |     |

Intercept was enabled in Burp Suite before navigating to the DVWA login page. When the login form was submitted with test credentials, the request was captured by Burp before reaching the server. The intercepted request showed a POST method containing the username and password in the request body, along with cookies and session data. This demonstrated that login credentials are visible in intercepted traffic

at the proxy level before being processed by the server.

## TASK 8: Forward and See Response

The screenshot displays the Burp Suite interface. At the top, the 'HTTP history' tab is active, showing a list of intercepted requests. The table has columns for #, Host, Method, URL, Params, Edited, Status code, Length, MIME type, Extension, Title, Notes, TLS, IP, and Cool. The selected request is #32, a GET request to http://127.0.0.1/dvwa/login.php, with a status code of 302 and a length of 295. Below the table, the 'Request' and 'Response' panels are visible. The 'Request' panel shows the raw HTTP request, including headers like 'Host: 127.0.0.1', 'Cache-Control: max-age=0', 'sec-ch-ua: "Not-A.Brand";v="24", "Chromium";v="146"', 'sec-ch-ua-mobile: ?0', 'sec-ch-ua-platform: "Linux"', 'Accept-Language: en-US,en;q=0.9', 'Upgrade-Insecure-Requests: 1', 'User-Agent: Mozilla/5.0 (X11; Linux x86\_64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/146.0.0.0 Safari/537.36', 'Accept: text/html,application/xhtml+xml,application/xml;q=0.9,image/avif,image/webp,image/apng,\*/\*;q=0.8,application/signed-exchange;v=b3;q=0.7', 'Sec-Fetch-Site: same-origin', 'Sec-Fetch-Mode: navigate', 'Sec-Fetch-User: ?1', 'Sec-Fetch-Dest: document', 'Referer: http://127.0.0.1/dvwa/login.php', 'Accept-Encoding: gzip, deflate, br', 'Cookie: security=impossible; PHPSESSID=4fcc69ae9aaafb199a3b77a867e3f7c8', and 'Connection: keep-alive'. The 'Response' panel shows the raw HTTP response, including headers like 'HTTP/1.1 200 OK', 'Date: Sun, 07 Jun 2026 16:37:12 GMT', 'Server: Apache/2.4.66 (Debian)', 'Expires: Tue, 23 Jun 2009 12:00:00 GMT', 'Cache-Control: no-cache, must-revalidate', 'Pragma: no-cache', 'Vary: Accept-Encoding', 'Content-Length: 6443', 'Keep-Alive: timeout=5, max=100', 'Connection: Keep-Alive', and 'Content-Type: text/html; charset=utf-8'. The response body is an HTML document with a title 'Welcome :: Damn Vulnerable Web Application (DVWA)' and links to stylesheets and scripts.

| #  | Host             | Method | URL             | Params | Edited | Status code | Length | MIME type | Extension | Title                      | Notes | TLS | IP        | Cool |
|----|------------------|--------|-----------------|--------|--------|-------------|--------|-----------|-----------|----------------------------|-------|-----|-----------|------|
| 29 | http://127.0.0.1 | GET    | /dvwa/          |        |        | 302         | 337    | HTML      |           |                            |       |     | 127.0.0.1 |      |
| 30 | http://127.0.0.1 | GET    | /dvwa/login.php |        |        | 500         | 295    | HTML      | php       |                            |       |     | 127.0.0.1 |      |
| 31 | http://127.0.0.1 | GET    | /dvwa/          |        |        | 302         | 337    | HTML      |           |                            |       |     | 127.0.0.1 |      |
| 32 | http://127.0.0.1 | GET    | /dvwa/login.php |        |        | 500         | 295    | HTML      | php       |                            |       |     | 127.0.0.1 |      |
| 33 | http://127.0.0.1 | GET    | /dvwa/          |        |        | 302         | 337    | HTML      |           |                            |       |     | 127.0.0.1 |      |
| 34 | http://127.0.0.1 | GET    | /dvwa/login.php |        |        | 500         | 295    | HTML      | php       |                            |       |     | 127.0.0.1 |      |
| 35 | http://127.0.0.1 | GET    | /dvwa/setup.php |        |        | 200         | 5047   | HTML      | php       | Setup :: Damn Vulnerabl... |       |     | 127.0.0.1 |      |
| 43 | http://127.0.0.1 | POST   | /dvwa/setup.php |        | ✓      | 302         | 343    | HTML      | php       |                            |       |     | 127.0.0.1 |      |
| 44 | http://127.0.0.1 | GET    | /dvwa/setup.php |        |        | 200         | 5948   | HTML      | php       | Setup :: Damn Vulnerabl... |       |     | 127.0.0.1 |      |

**Request**

Pretty Raw Hex

```
1 GET /dvwa/index.php HTTP/1.1
2 Host: 127.0.0.1
3 Cache-Control: max-age=0
4 sec-ch-ua: "Not-A.Brand";v="24", "Chromium";v="146"
5 sec-ch-ua-mobile: ?0
6 sec-ch-ua-platform: "Linux"
7 Accept-Language: en-US,en;q=0.9
8 Upgrade-Insecure-Requests: 1
9 User-Agent: Mozilla/5.0 (X11; Linux x86_64) AppleWebKit/537.36 (KHTML, like Gecko)
10 Chrome/146.0.0.0 Safari/537.36
11 Accept:
12 text/html,application/xhtml+xml,application/xml;q=0.9,image/avif,image/webp,image/ap
13 ng,*/*;q=0.8,application/signed-exchange;v=b3;q=0.7
14 Sec-Fetch-Site: same-origin
15 Sec-Fetch-Mode: navigate
16 Sec-Fetch-User: ?1
17 Sec-Fetch-Dest: document
18 Referer: http://127.0.0.1/dvwa/login.php
19 Accept-Encoding: gzip, deflate, br
20 Cookie: security=impossible; PHPSESSID=4fcc69ae9aaafb199a3b77a867e3f7c8
21 Connection: keep-alive
```

**Response**

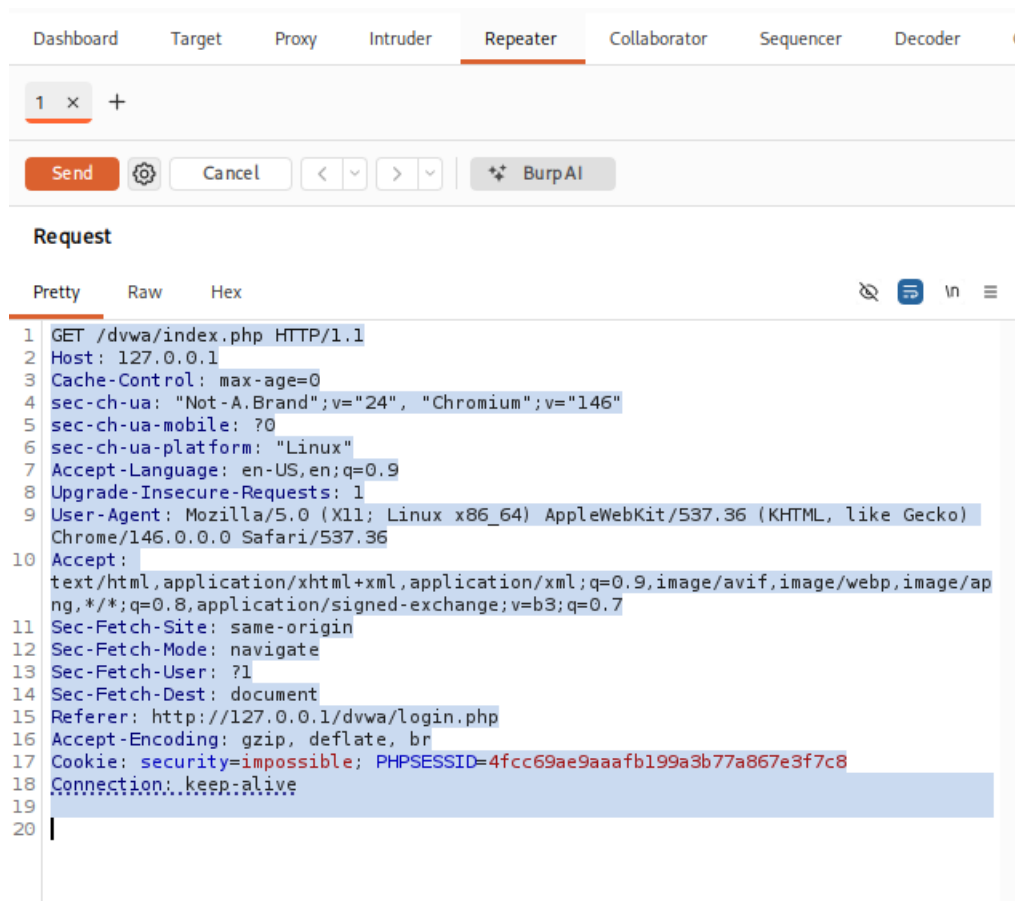
Pretty Raw Hex Render

```
1 HTTP/1.1 200 OK
2 Date: Sun, 07 Jun 2026 16:37:12 GMT
3 Server: Apache/2.4.66 (Debian)
4 Expires: Tue, 23 Jun 2009 12:00:00 GMT
5 Cache-Control: no-cache, must-revalidate
6 Pragma: no-cache
7 Vary: Accept-Encoding
8 Content-Length: 6443
9 Keep-Alive: timeout=5, max=100
10 Connection: Keep-Alive
11 Content-Type: text/html; charset=utf-8
12
13 <!DOCTYPE html>
14
15 <html lang="en-GB">
16
17 <head>
18 <meta http-equiv="Content-Type" content="text/html; charset=UTF-8" />
19
20 <title>
21 Welcome :: Damn Vulnerable Web Application (DVWA)
22 </title>
23
24 <link rel="stylesheet" type="text/css" href="dvwa/css/main.css" />
25
26 <link rel="icon" type="image/ico" href="favicon.ico" />
27
28 <script type="text/javascript" src="dvwa/js/dvwaPage.js">
29
30 </script>
31
32 </head>
```

After intercepting the login request, the Forward button was used to send the request to the server. Burp then displayed the server's response, which was also forwarded back to the browser. The browser successfully loaded the DVWA page after authentication, confirming that the login request had been processed correctly. This step showed how Burp allows inspection of both request and response before they reach the browser.

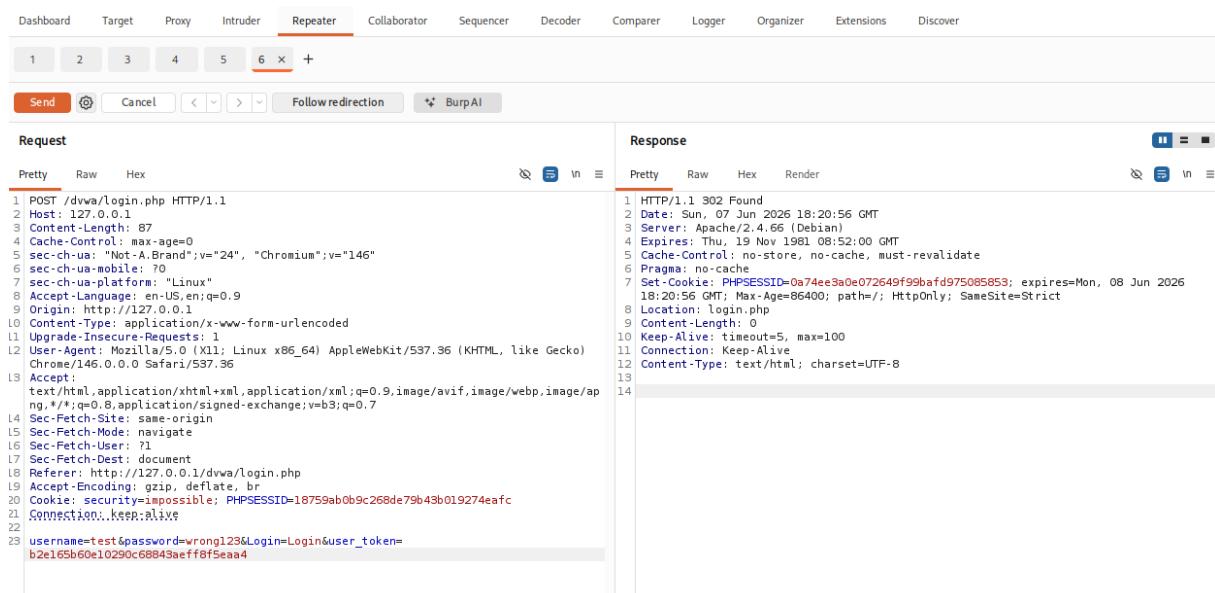
## TASK 9a: Find a Request in HTTP History

## TASK 9b: Send to Repeater



A previously captured HTTP request from the Proxy HTTP History was selected and sent to the Repeater tool. This action copied the request into a controlled testing environment where it could be modified and resent multiple times without using the browser. The request was loaded into Repeater with all headers, cookies, and parameters preserved. This allowed manual manipulation of the request before sending it again to observe how the server reacts to modified input.

## TASK 10: Change a Parameter



Inside the Repeater tool, a parameter in the HTTP request (such as username,

password, or input value) was modified manually. The altered request was then sent to the server using the Send button. The server responded with a new HTTP response, such as a redirect (302 Found) or updated page content, depending on the modified input. This demonstrated how changes in request parameters directly affect server behaviour and allowed observation of how input validation and authentication handling are implemented on the server side.

## TASK 11: Look at Response Details

The screenshot shows the Burp Suite interface. The top bar includes tabs for Intercept, HTTP history, WebSockets history, Match and replace, and Proxy settings. Below this is a filter bar with the text 'Filter settings: Hiding CSS and image content; hiding specific extensions'. The main table displays a list of HTTP requests and responses. The columns include #, Host, Method, URL, Params, Edited, Status code, Length, MIME type, Extension, Title, Notes, TLS, IP, and Cookies. A request to 'http://127.0.0.1/dvwa/login.php' is selected, showing a 200 OK status. Below the table, the 'Request' and 'Response' sections are visible. The 'Request' section shows the raw HTTP request, including headers like 'Host: 127.0.0.1', 'sec-ch-ua: "Not-A.Brand";v="24", "Chromium";v="146"', and 'User-Agent: Mozilla/5.0 (X11; Linux x86\_64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/146.0.0.0 Safari/537.36'. The 'Response' section shows the raw HTTP response, including headers like 'HTTP/1.1 200 OK', 'Date: Sun, 07 Jun 2026 16:20:14 GMT', 'Server: Apache/2.4.66 (Debian)', and 'Content-Type: text/html; charset=utf-8'. The response body is shown as a Base64 encoded string.

| #  | Host                       | Method | URL                | Params | Edited | Status code | Length | MIME type | Extension | Title                      | Notes | TLS | IP           | Coc |
|----|----------------------------|--------|--------------------|--------|--------|-------------|--------|-----------|-----------|----------------------------|-------|-----|--------------|-----|
| 33 | http://127.0.0.1           | GET    | /dvwa/             |        |        | 302         | 337    | HTML      |           |                            |       |     | 127.0.0.1    |     |
| 34 | http://127.0.0.1           | GET    | /dvwa/login.php    |        |        | 500         | 295    | HTML      | php       |                            |       |     | 127.0.0.1    |     |
| 35 | http://127.0.0.1           | GET    | /dvwa/setup.php    |        |        | 200         | 5047   | HTML      | php       | Setup :: Damn Vulnerabl... |       |     | 127.0.0.1    |     |
| 43 | http://127.0.0.1           | POST   | /dvwa/setup.php    |        | ✓      | 302         | 343    | HTML      | php       |                            |       |     | 127.0.0.1    |     |
| 44 | http://127.0.0.1           | GET    | /dvwa/setup.php    |        |        | 200         | 5948   | HTML      | php       | Setup :: Damn Vulnerabl... |       |     | 127.0.0.1    |     |
| 45 | http://127.0.0.1           | GET    | /dvwa/login.php    |        |        | 302         | 343    | HTML      | php       |                            |       |     | 127.0.0.1    |     |
| 47 | http://127.0.0.1           | POST   | /dvwa/setup.php    |        | ✓      | 302         | 343    | HTML      | php       |                            |       |     | 127.0.0.1    |     |
| 48 | http://127.0.0.1           | GET    | /dvwa/setup.php    |        |        | 200         | 5948   | HTML      | php       | Setup :: Damn Vulnerabl... |       |     | 127.0.0.1    |     |
| 49 | http://127.0.0.1           | GET    | /dvwa/login.php    |        |        | 302         | 337    | HTML      |           |                            |       |     | 127.0.0.1    |     |
| 50 | http://127.0.0.1           | GET    | /dvwa/             |        |        | 200         | 1669   | HTML      | php       | Login :: Damn Vulnerabl... |       |     | 127.0.0.1    |     |
| 54 | http://127.0.0.1           | POST   | /dvwa/login.php    |        | ✓      | 302         | 482    | HTML      | php       |                            |       |     | 127.0.0.1    | PHI |
| 55 | http://127.0.0.1           | GET    | /dvwa/index.php    |        |        | 200         | 6857   | HTML      | php       | Welcome :: Damn Vulner...  |       |     | 127.0.0.1    |     |
| 56 | http://127.0.0.1           | GET    | /dvwa/security.php |        |        | 200         | 5516   | HTML      | php       | DVWA Security :: Damn ...  |       |     | 127.0.0.1    |     |
| 58 | http://testphp.vulnweb.com | GET    | /login.php         |        |        |             |        | HTML      | php       |                            |       |     | 44.228.249.3 |     |
| 59 | http://testphp.vulnweb.com | GET    | /login.php         |        |        |             |        | HTML      | php       |                            |       |     | 44.228.249.3 |     |
| 60 | http://testphp.vulnweb.com | GET    | /login.php         |        |        |             |        | HTML      | php       |                            |       |     | 44.228.249.3 |     |
| 62 | http://testphp.vulnweb.com | GET    | /login.php         |        |        |             |        | HTML      | php       |                            |       |     | 44.228.249.3 |     |
| 64 | http://testphp.vulnweb.com | GET    | /login.php         |        |        |             |        | HTML      | php       |                            |       |     | 44.228.249.3 |     |
| 66 | http://127.0.0.1           | GET    | /dvwa/login.php    |        |        | 200         | 1670   | HTML      | php       | Login :: Damn Vulnerabl... |       |     | 127.0.0.1    |     |
| 67 | http://127.0.0.1           | GET    | /dvwa/login.php    |        |        | 200         | 1670   | HTML      | php       | Login :: Damn Vulnerabl... |       |     | 127.0.0.1    |     |
| 68 | http://127.0.0.1           | POST   | /dvwa/login.php    |        | ✓      | 302         | 482    | HTML      | php       |                            |       |     | 127.0.0.1    | PHI |
| 69 | http://127.0.0.1           | GET    | /dvwa/index.php    |        |        | 200         | 6858   | HTML      | php       | Welcome :: Damn Vulner...  |       |     | 127.0.0.1    |     |
| 70 | http://127.0.0.1           | GET    | /dvwa/index.php    |        |        | 200         | 6771   | HTML      | php       | Welcome :: Damn Vulner...  |       |     | 127.0.0.1    |     |
| 71 | http://127.0.0.1           | GET    | /dvwa/index.php    |        |        | 200         | 6771   | HTML      | php       | Welcome :: Damn Vulner...  |       |     | 127.0.0.1    |     |

**Request**

Pretty Raw Hex

```
1 GET /dvwa/Login.php HTTP/1.1
2 Host: 127.0.0.1
3 sec-ch-ua: "Not-A.Brand";v="24", "Chromium";v="146"
4 sec-ch-ua-mobile: ?0
5 sec-ch-ua-platform: "Linux"
6 Accept-Language: en-US,en;q=0.9
7 Upgrade-Insecure-Requests: 1
8 User-Agent: Mozilla/5.0 (X11; Linux x86_64) AppleWebKit/537.36 (KHTML, like
9 Gecko) Chrome/146.0.0.0 Safari/537.36
10 Accept:
11 text/html,application/xhtml+xml,application/xml;q=0.9,image/avif,image/webp,ima
12 ge/apng,*/*;q=0.8,application/signed-exchange;v=b3;q=0.7
13 Sec-Fetch-Site: none
14 Sec-Fetch-Mode: navigate
```

**Response**

Pretty Raw Hex Render

```
1 HTTP/1.1 200 OK
2 Date: Sun, 07 Jun 2026 16:20:14 GMT
3 Server: Apache/2.4.66 (Debian)
4 Expires: Tue, 29 Jun 2009 12:00:00 GMT
5 Cache-Control: no-cache, must-revalidate
6 Pragma: no-cache
7 Vary: Accept-Encoding
8 Content-Length: 1342
9 Keep-Alive: timeout=5, max=100
10 Connection: Keep-Alive
11 Content-Type: text/html; charset=utf-8
12
13 <!DOCTYPE html>
14
```

HTTP History was used to inspect captured requests and their corresponding responses. A request to the DVWA login page was selected, and the response showed a 200 OK status code along with HTTP headers such as Content-Type and Cache-Control. The response body contained the full HTML structure of the login page, including form fields and hidden tokens. This demonstrated how Burp reveals full server responses, including headers and page content returned to the browser.

## TASK 12b: Decode Using Burp Decoder

The screenshot shows the Burp Suite Decoder tool. The top bar includes tabs for Dashboard, Target, Proxy, Intruder, Repeater, Collaborator, Sequencer, Decoder, Comparer, Logger, Organizer, Extensions, and Discover. The Decoder tab is selected. The main area shows a Base64 encoded string: 'dXNlcm5hbWU6cGZc3dvcnQ='. Below this, the decoded string is shown: 'username:password'.

Dashboard Target Proxy Intruder Repeater Collaborator Sequencer **Decoder** Comparer Logger Organizer Extensions Discover

dXNlcm5hbWU6cGZc3dvcnQ=

username:password

The Decoder tool in Burp Suite was used to decode a Base64 encoded



authentication string. The string dXNlcm5hbWU6cGFzc3dvcmQ= was entered into the Decoder and processed using Base64 decoding. The output displayed the original credentials as username:password. This showed how encoded authentication data can be easily reversed using Burp Decoder to reveal readable credentials.